Experimental Gerontology 156 (2021) 111593

Contents lists available at ScienceDirect

Experimental Gerontology

journal homepage: www.elsevier.com/locate/expgero

Review

High-velocity resistance training as a tool to improve functional performance and muscle power in older adults

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ARTICLE INFO

Section Editor: Li-Ning Peng

Keywords:
Exercise
Resistance training
Power training
Function
Aged

ABSTRACT

The aging process results in several physiological impairments that, in turn, may predispose older individuals to a series of restrictions on their functional capacity. These impairments are important to understand so that suitable conditions for healthier aging can be pursued. In this review, we first summarize the effects of aging on the neuromuscular system, as well as on the relation between the main age-associated physiological impairments and functional performance with an emphasis on muscle power output. We then proceed to discuss the effects of resistance training, specifically high-velocity resistance training (HVRT), on the aforementioned neuromuscular impairments, and on functional performance in healthy and mobility-limited older adults. Collectively, available evidence suggests that HVRT seems to be a safe and effective intervention for improving muscle power, functional performance, and mobility of older individuals. It also seems that mobility-limited older adults may improve power and functional performance to a greater extent than their healthy counterparts after HVRT, which is in line with the principle of diminishing returns. Considering that only a very limited number of investigations directly compared the effects of HVRT in more than one of the aforementioned groups, studies comparing the adaptations to HVRT of middle-aged adults and older adults with distinct functional capacities would be valuable to determine whether there are differences in neuromuscular adaptations, functional performance, and functional reserve among these groups.

1. Introduction

The world's population aging rate is increasing in both absolute and relative terms (World Health Organization, 2015). According to estimates from the World Health Organization, in 2015 there were approximately 900 million people aged 60 and over, and that number is expected to reach the 2 billion mark by 2050 (World Health Organization, 2015). Except for the African continent, this means approximately a quarter of the remaining worldwide population will be over 60 years old. Far beyond a simple demographic transition, population aging results in a heavy economic and health burden on society (Janssen et al., 2004), as aging is an important risk factor for the development of chronic, non-communicable diseases, which are collectively responsible for almost 70% of deaths worldwide (World Health Organization, 2013). More recently, the coronavirus pandemic posed a major threat to older individuals and those with chronic diseases, considered as a high-risk group and most vulnerable to the disease (Williamson et al., 2020). In all these cases, the increase in risk stems from a myriad of physiological processes that occur during aging. Ultimately, these processes are important to understand so that suitable conditions for healthier aging can be pursued. In this review, we focus on neuromuscular aging and functional consequences. We first summarize the effects of aging on the neuromuscular system, and the relation between age-associated physiological impairments and functional performance with an emphasis on muscle power output. We then discuss the effects of resistance training, specifically high-velocity resistance training (HVRT), on the aforementioned neuromuscular impairments and functional performance in healthy and mobility-limited older adults.

2. Aging and its effects on the neuromuscular system

Aging can, from a biological perspective, be defined as the result of the cumulative impact of a wide variety of molecular and cellular damage over time (Fragala, 2015a; World Health Organization, 2015).

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https://doi.org/10.1016/j.exger.2021.111593
Received 22 August 2021; Received in revised form 28 September 2021; Accepted 8 October 2021
Available online 14 October 2021
0531-5565/© 2021 Published by Elsevier Inc.

This process results in the natural deterioration of several body systems, such as the neuromuscular, cardiovascular, respiratory and bone systems (Hunter et al., 2016; Jakovljevic, 2018; Powers and Howley, 2018; Roberts et al., 2016; Roman et al., 2016; van den Beld et al., 2018). Although relevant, addressing all these systems is outside the scope of the present review and, therefore, only the main physiological changes related to the neuromuscular system are highlighted.

From the fifth decade of life and more significantly from the age of 60 onwards there is a considerable loss of muscle mass (Aoyagi and Shephard, 1992), classically termed sarcopenia (Rosenberg, 1989). This reduction seems to be linked to decreases in both the total number of muscle fibers (Lexell et al., 1983, 1986, 1988; Roberts et al., 2018), and the size of type II fibers (Hepple, 2018; Hepple and Rice, 2016; Nilwik et al., 2013; Piasecki et al., 2018; Roberts et al., 2018). The reduction in muscle strength is also a hallmark of aging, being considered for a long time as a direct result of the loss of muscle mass. Nevertheless, it is now well established that the annual reduction in muscle strength is more pronounced than that of muscle mass (~3% vs. 1%) (Goodpaster et al., 2006; Mitchell et al., 2012), indicating age-related declines in muscle strength are multifactorial and only partially driven by muscle atrophy.

In fact, the complex etiology, with overlapping and interdependent mechanisms, hinders a definitive understanding of the processes behind the losses of muscle mass and strength. Yet, some mechanisms are suggested to be important in the development of these phenotypes, such as neural changes and reductions in protein synthesis (Dobrowolny et al., 2021; Graham et al., 2021; Lavin et al., 2019). The progressive loss of alpha motoneurons leads to denervation of the fibers innervated by it and predisposes them to apoptosis. In order to combat these events, aging muscle goes through extensive motor unit remodeling where adjacent motor neurons begin to innervate these fibers, expanding the size of individual motor units (Piasecki et al., 2018). The success of this process, known as the denervation-reinnervation cycle, contributes to the reduction of atrophy and progression of sarcopenia. However, at older ages there seems to be a compromise in this reinnervation process (Deschenes, 2011; Dobrowolny et al., 2021; Kung et al., 2014; Mitchell et al., 2012; Piasecki et al., 2018). Even when successful, denervation-related motor unit remodeling also tends to lead to the clustering of type I fibers, known as myofiber grouping, which is related to disrupted motor unit recruitment (Kelly et al., 2018). The loss of motor neurons also lowers the number of motor units; thus a greater proportion of myofibers will be activated at any given intensity (Graham et al., 2021).

Furthermore, there also appears to be a reduction in neuromuscular activation of agonist muscles (Kelly et al., 2018; Reid et al., 2014; Reid and Fielding, 2012). That is, an impairment in the process by which the nervous system produces muscle strength through the recruitment and firing frequency of the motor units, often accompanied by an increase in co-activation of the antagonistic muscles (Clark et al., 2010; Clark and Fielding, 2012; Häkkinen et al., 1998a). As for the muscle itself, there seems to be a reduction in muscle quality, usually related to an increase in the infiltration of fat and non-contractile fibrous tissue in the skeletal muscle, which is negatively associated with strength production and movement velocity in older individuals (Fukumoto et al., 2012). For these and countless other factors, what is observed is a slower contractile phenotype in these individuals (Leyva et al., 2016; Reid et al., 2014). Accordingly, previous studies have shown a reduction in movement velocity in older versus younger adults (De Vito et al., 1998; Leyva et al., 2016; Raj et al., 2010) and in healthy versus mobility-limited older adults (Reid et al., 2014).

Considering its direct relationship with strength and movement velocity, it is expected that muscle power (i.e., the amount of work done in a given time interval) will also be affected. In fact, both cross-sectional and longitudinal studies have observed lower muscle power outputs in older individuals (Izquierdo et al., 1999; Leyva et al., 2016; McNeil et al., 2007; Metter et al., 1997; Reid et al., 2014; Roberts et al., 2018; Skelton et al., 1994). In addition, when compared to strength, power seems to decline earlier and at higher rates (~2× more) with advancing age (Kennis et al., 2014; Skelton et al., 1994). Analysis of knee extension power output in 357 men and women aged 18 to 80 years identified that reductions begin between the fourth and fifth decade of life and are more severe when analyzed at higher movement velocities (Leyva et al., 2016). A recent longitudinal study (Kennis et al., 2014) measured isometric and dynamic strength, muscle quality and power output of 105 men aged between 45 and 49 years, who were assessed again 10 years later. The authors identified a reduction in strength measurements close to 1% per year and almost 2% in power output and muscle quality, supporting the results observed by Leyva et al. (Leyva et al., 2016).

Collectively, these and other physiological and structural changes that occur during aging impact, for example, the performance of activities of daily living and the mobility of these older individuals (Reid and Fielding, 2012; Visser et al., 2005). These impairments can also lead to limitations, impairments in functional performance, loss of independence, disability and, eventually, death (e.g., Brown et al., 2016; Guralnik et al., 1995; Newman et al., 2006; Srikanthan and Karlamangla, 2014). The main challenge is, therefore, to make these losses "*as close as possible to rectangular so that the function is maintained to the extent possible throughout the lifespan*" (Rosenberg, 1989).

3. Neuromuscular aging and functional performance

The disablement process model suggests that both acute and chronic conditions lead to impairment or dysfunction in specific body systems. These impairments lead to limitations in the functions of the system or organism and, consequently, preclude it to act in the necessary or expected way (Verbrugge and Jette, 1994). Through this prism, we can view the aging process as a condition that results in several physiological impairments that, in turn, predispose older individuals to a series of restrictions on their functional capacity (Fig. 1). Thus, the ability to perform activities such as climbing stairs, getting up from a chair or walking becomes impaired until, in some cases, they are no longer able to perform them (Reid and Fielding, 2012). Maintenance of functional performance is, therefore, important for the preservation of mobility, independence and performance of activities of daily living (Guralnik et al., 2000).

3.1. Muscle mass and strength

Based on this premise several studies in the last decades have sought to identify the main physiological impairments most strongly associated with functional performance in older individuals (Byrne et al., 2016). Associations between muscle mass and functional performance were observed (Casas-Herrero et al., 2013; Newman et al., 2003; Reid et al., 2008; Visser et al., 2002), however, there is evidence to suggest that this association appears to be, at least in part, dependent or mediated by muscle strength in the healthy elderly population (Visser et al., 2005). Likewise, meta-analysis of prospective studies did not find a significant relationship between lower muscle mass and higher risk of functional decline in adults over 55 years (Schaap et al., 2013). On the other hand, lower muscle strength significantly increased the risk of functional decline (OR = 1.86; 95% CI 1.32-2.64). In fact, the impact of muscle strength loss and its association with functional capacity seems well established (Bassey et al., 1988; Bean et al., 2003, 2002; Brown et al., 1995; Buchner et al., 1996; Casas-Herrero et al., 2013; Fiatarone et al., 1990; Lord et al., 2002).

3.2. Muscle power

More recently, muscle power has been suggested as a determining factor for functional performance in this population (Reid and Fielding, 2012). As highlighted before, although strongly associated with strength, both do not represent the same phenomenon (Bean et al., 2002; Sayers, 2007), considering that power is determined by strength (i.e., the ability to exert force to overcome resistance), but also by movement

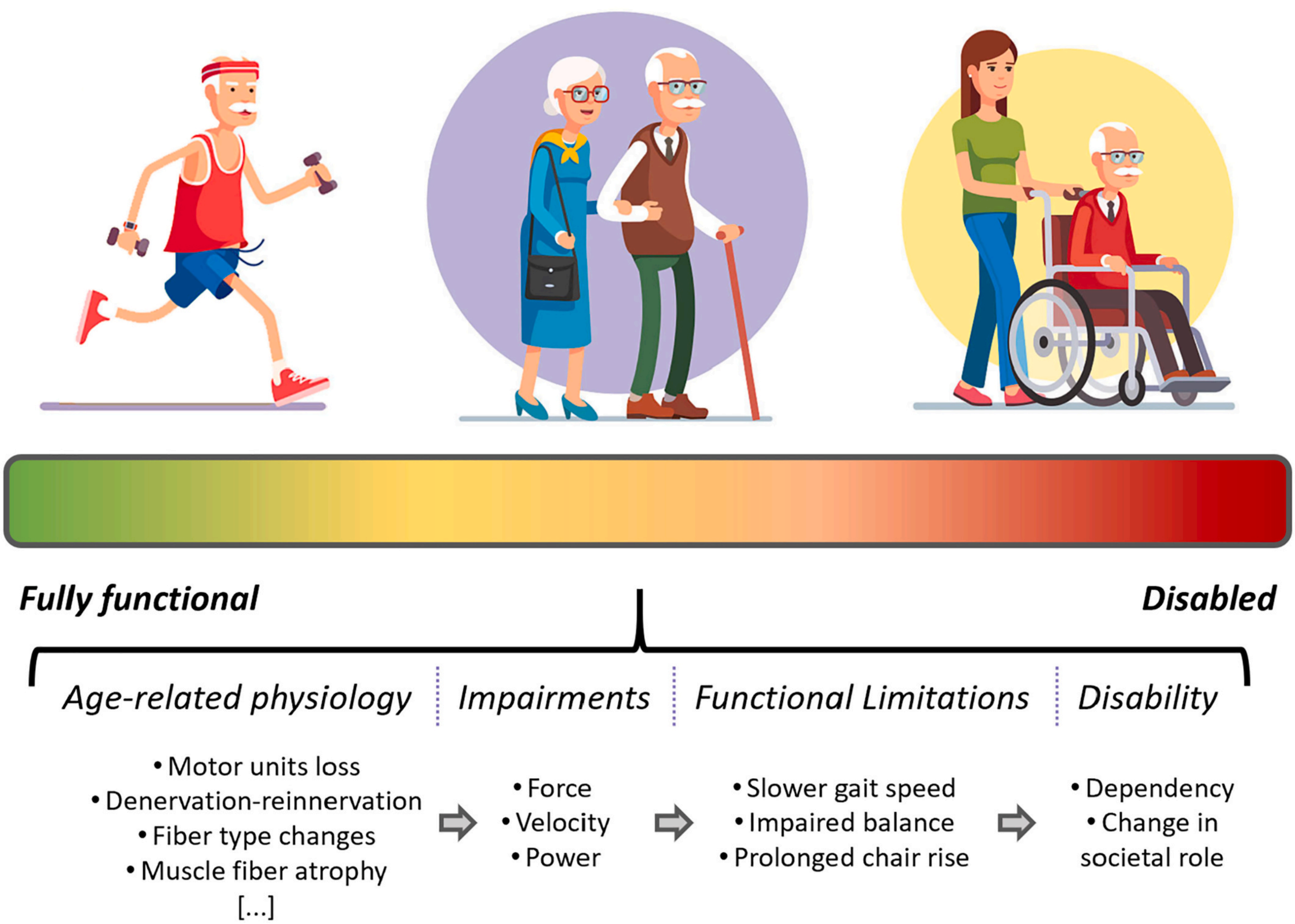

Fig. 1. Disablement process model applied to aging (adapted from REID; FIELDING, 2012 with permission).

velocity (i.e., power = force × velocity). The relationship between power and functional performance is observed in different groups of older individuals, from those apparently healthy (Kozakai et al., 2000; Larsen et al., 2009; Petrella et al., 2004; Samson et al., 2000; Skelton et al., 1994) and with limited mobility (Bean et al., 2002; Foldvari et al., 2000; Suzuki et al., 2001) to those institutionalized and frail (Bassey et al., 1992; Casas-Herrero et al., 2013; Clémençon et al., 2008). The reduction in muscle power with aging is also associated with a reduction in mobility (Bassey et al., 1992; Bean et al., 2003, 2002) and an increase in the incidence of falls (Casas-Herrero et al., 2013; Skelton et al., 2002).

Among the first studies to investigate this relationship in healthy older adults, Skelton et al. (Skelton et al., 1994) identified negative correlations between lower limb muscle power and chair rise time ($r = -0.38$ to -0.56) in elderly men and women (65–89 years). Similar results had also been observed in institutionalized older adults (80–99 years) for the sit-to-stand ($r = 0.65$), stair climb ($r = 0.81$) and habitual gait speed tests ($r = 0.80$) (Bassey et al., 1992). The difference in the direction of the associations between the two studies is because while Skelton al. used the time to perform the tests, Bassey et al. used the actual speed to get up from the chair. These results were confirmed and positive associations were also observed between lower limb muscle power and gait speed ($r = 0.38$–0.62) in middle-aged and older adults (Casas-Herrero et al., 2013; Kozakai et al., 2000; Rantanen and Avela, 1997) and in different functional tests, such as the *Short Physical Performance Battery* (SPPB) (Bean et al., 2003, 2002), sit-to-stand (Carabello et al., 2010; Casas-Herrero et al., 2013; Suzuki et al., 2001), stair climb (Larsen et al., 2009; Suzuki et al., 2001), timed up and go (TUG) (Stenroth et al., 2015) and the 6-minute or 400 m walk tests (Marsh et al., 2006; Stenroth et al., 2015).

3.3. *Muscle power or strength?*

Clearly, both strength and power are important predictors of functional performance in these individuals (Byrne et al., 2016), however, is it possible that one is more associated than the other? The current literature seems to suggest that power is in fact more strongly associated with functional capacity than muscle strength (ACSM, 2009). Furthermore, power seems to be a better predictor of performance in tasks related to mobility and ambulation capacity (Bassey et al., 1992; Bean et al., 2003, 2002; Cuoco et al., 2004; Foldvari et al., 2000; Reid et al., 2014; Suzuki et al., 2001), explaining between 2 and 8% more of the variability in functional performance measured in a range of different tests (Bean et al., 2003), including in mobility-limited older adults (Bean et al., 2002). Corroborating these results, Cuoco et al. (Cuoco et al., 2004) observed higher coefficients of determination between power at 40% and 70% of 1RM and sit-to-stand ($R^2 = 0.12$–0.15 vs. 0.09), stair climb ($R^2 = 0.14$–0.18 vs. 0.10) and gait speed tests performance ($R^2 = 0.26$–0.35 vs. 0.07) when compared to muscle strength. Moreover, lower power levels are associated with a three times greater risk of impaired mobility than the risk related to lower muscle strength (Bean et al., 2010) and it is possible to observe an improvement in physical functioning due to an increase in muscle power independent of changes in strength (Bean et al., 2010, 2002).

3.4. But which muscle power?

Although the importance of muscle power for functional performance is well established in the literature (Byrne et al., 2016; Orssatto et al., 2018; Reid and Fielding, 2012), the association between these two outcomes is likely influenced by different factors, such as the relative load at which power is measured. As an example, even though power measured at 40% and 70% of the one repetition maximum (1RM) load explained a portion of the variance in both the sit-to-stand ($R^2 = 0.28$ vs. 0.24) and stair climb ($R^2 = 0.42$ vs. 0.43) tests to a similar degree, power output at 40% explained a greater proportion of the variance in gait speed ($R^2 = 0.59$ vs. 0.51) when compared to the 70% load (Cuoco et al., 2004). A similar result was observed by Puthoff and Nielsen (Puthoff and Nielsen, 2007). Specifically, these authors observed that power at 40% 1RM explained a higher proportion of the variance in the gait speed test ($R^2 = 0.31$ vs. 0.24), while power at 90% explained a higher proportion of variance in the sit-to-stand test ($R^2 = 0.20$ vs. 0.34). But why?

A possible explanation may be related to the fact that different functional tasks, such as raising from a chair, walking down a hallway or quickly repositioning the lower limb after losing balance likely depend on different contributions from muscle power determinants, that is, strength and velocity (Byrne et al., 2016). The action of raising from a chair, for example, seems to require a higher percentage of the maximum knee extension torque (Hortobágyi et al., 2003), while the relative effort during walking has a lower demand (~30%) (Beijersbergen et al., 2013). Recovery after an imbalance or the act of quickly crossing an intersection also require lower percentages of maximal strength, but which need to be produced quickly (Sayers, 2007).

In addition to differences between functional tasks, aging may also predispose individuals to rely more or less on these aforementioned parameters. This notion was experimentally verified in middle-aged adults and older individuals with and without mobility limitations (Pojednic et al., 2012) by cross-sectionally looking into the inter-individual variation in muscle power that was explained by strength and movement velocity. It was shown that only strength significantly explained the variance in muscle power in the middle-aged participants (partial $R^2 = 0.30$), whereas both strength and velocity contributed in healthy older adults (partial $R^2 = 0.36$ and 0.40) and only velocity explained muscle power in those with mobility limitations (partial $R^2 = 0.13$), suggesting that the importance of movement velocity for power production likely increases with advancing age or as functional performance decreases. More recently, Alcazar et al. (2018) divided a group of older adults aged 70 or over into strength and velocity tertiles. Those in the lowest strength or velocity tertiles had significantly lower power values compared to the remaining participants, but not between them, suggesting that the lower power output may occur due to deficits in strength for some older individuals and due to deficits in velocity in others. In fact, Alcazar et al. proposed that the results observed by Pojednic et al. (2012) could indicate a higher prevalence of individuals with velocity deficits in the latter mobility-limited older adults' sample.

3.5. To muscle power and beyond (it)

It is important to note that, although muscle power explains between one third and half of the variance in several functional performance outcomes (Byrne et al., 2016), a portion of this variance still remains unexplained. Thus, other impairments related to the aging process, such as reductions in neuromuscular activation and muscle quality, also seem to be associated with functional performance in these individuals and could possibly explain an additional portion of this remaining variance (Clark et al., 2011, 2013; Lopez et al., 2017; Rech et al., 2014; Reid et al., 2014; Wilhelm et al., 2014).

According to Clark et al. (2011), few studies to date have investigated the association between impairments in neuromuscular activation and functional decline during aging. Among the studies identified, neuromuscular activation of the quadriceps muscles seems to be positively associated ($r = 0.56$) with SPPB scores in a joint analysis of older adults with and without mobility limitations (Clark et al., 2011), whereas significant associations were observed between the rate of increase in neuromuscular activation of the medial gastrocnemius and maximal gait speed ($r = 0.54$) in healthy older adults (Clark et al., 2013). Quadriceps neuromuscular activation also seems capable of differentiating older women with higher or lower habitual gait speed (Brach et al., 2001). Recently, however, we were unable to identify an association between quadriceps neuromuscular activation and SPPB performance in a sample of middle-aged and older adults with and without limitation (*unpublished data*), and even though neuromuscular activation was associated with sit-to-stand performance, both strength and power were found to be more associated with the former ($r = 0.30$ vs. $r = 0.51$ to 0.66).

As for muscle quality, the literature seems to agree that it is reduced in older individuals when compared to their younger counterparts (Fragala et al., 2015; Straight et al., 2013a). That is true even though there is currently no consensus definition of muscle quality, so that different techniques and measures are applied (Fragala et al., 2015). Nevertheless, reductions in muscle quality are commonly related to an increase in intramuscular fat and connective tissue infiltration (Arts et al., 2010) and are also associated with an impairment in the capacity to produce both strength and power (Cadore et al., 2012; Fragala et al., 2015; Fukumoto et al., 2012; Straight et al., 2013a, 2013b; Wilhelm et al., 2014). When it comes to functional performance, previous studies demonstrated significant correlations between quadriceps muscle quality, measured through ultrasound echo intensity values, and the performance in the sit-to-stand test in elderly men and women ($r = -0.502$ to -0.493) (Rech et al., 2014; Wilhelm et al., 2014). The same research group observed later that the quadriceps echo intensity values were also independent predictors of functional performance, explaining close to 30% of the variance in the sit-to-stand test in sedentary older men (Lopez et al., 2017). Muscle quality analyses using computed tomography identified that lower signal attenuation values in the thigh muscles, indicative of lower density and greater intramuscular fat infiltration, were associated with worse performances in the gait speed and sit-to-stand tests in a sample of 3075 men and women aged 70 to 79 years (Visser et al., 2002). More importantly, these lower muscle quality results were also independent predictors of mobility limitations after a period of almost three years of follow-up in these same participants (Visser et al., 2005).

In summary, several physiological impairments are associated to a greater or lesser extent with the functional decline observed during aging. Of these, the reductions observed in power, whether through mechanisms related to strength or movement velocity, seem to have an important influence on the performance of tasks related to the daily life of the older population. This relationship also seems to be influenced by the age group analyzed (e.g., older compared to middle-aged adults), as well as by the functional capacity of these individuals (e.g., older adults with and without mobility limitations). Despite the recent advancements in the understanding of these associations, most investigations to date are restricted (i) to the comparison between a limited number of measures (e.g., strength vs. power or neuromuscular outcomes only), (ii) a small number of functional tests and (iii) evaluation of only a set of individuals (e.g., only healthy older adults or only mobility-limited older adults). Coupled with common methodological differences between studies (e.g., testing mode, exercise employed, %1RM in which the outcomes were measured, among others), it is difficult to compare and understand how these relationships behave between different aging and functional groups and, therefore, further investigation is required. A better understanding of these aspects can help guide the training and exercise prescription process, which will be covered in detail below.

4. Physical exercise in aging

Senescence can be classified into primary and secondary. The first involves the inevitable decline in cellular structure and function, while the second comprises the damage resulting from environmental and lifestyle exposures (Fragala, 2015b). Therefore, although it is not possible to prevent aging, it is evident that there are possibilities to intervene positively in the process. Physical exercise is a modifiable lifestyle factor and widely recognized as capable of improving several of the age-related physiological impairments discussed so far (ACSM, 2009; Byrne et al., 2016; Graham et al., 2021; Lavin et al., 2019; Paterson et al., 2007). Among the different exercise modes that have been investigated, traditional resistance training, i.e. that in which exercises are performed in a controlled manner (concentric and eccentric phases ~2 s each), is an effective and safe way to promote muscle hypertrophy and improve strength (Borde et al., 2015; Cadore et al., 2010; Cadore et al., 2014c; Radaelli et al., 2014), as well as power (Straight et al., 2016) and functional performance (Steib et al., 2010; Tschopp et al., 2011) in older individuals, including in those very old (~90 years) (Fiatarone et al., 1990). In fact, results from a study with 6242 individuals aged 18 to 98 years showed that those individuals who reported being regularly engaged in resistance exercises had a better performance on the sit-to-stand test when compared to sedentary individuals (Landi et al., 2018). Interestingly enough, results were also the same when those who reported being engaged in resistance exercises were compared to individuals who reported walking at their leisure time or who engaged regularly in aerobic exercise.

4.1. High-velocity resistance training

Currently, the main therapeutic strategies for improving functional capacity include some form of resistance exercise (Byrne et al., 2016; Cadore and Izquierdo, 2018). As previously discussed, power is strongly associated with functional performance and the maintenance of mobility, autonomy and independence in older people (Bean et al., 2010, 2002). It is not surprising, therefore, that there is an increase in attention aimed at identifying strategies capable of increasing muscle power among these individuals. As the responses to training are usually specific to the stimulus provided, it was suggested that performing the exercises in an explosive manner, that is, with the concentric portion of the exercises performed as quickly as possible, would favor muscle power adaptations (Bottaro et al., 2007; Reid and Fielding, 2012).

This form of resistance training, known as high-velocity resistance training (HVRT), is highly recommended for older adults (Fragala et al., 2019) and seems to result in greater muscle power and physical function improvements than traditional resistance training protocols in this population (Cadore et al., 2018; Cadore et al., 2014c; Reid and Fielding, 2012). This is demonstrated by both systematic reviews and meta-analyses (Byrne et al., 2016; Orssatto et al., 2018; Steib et al., 2010; Tschopp et al., 2011). Indeed, Byrne et al. (Byrne et al., 2016) observed that 10 of the 14 studies included in their systematic review supported HVRT as more effective than traditional resistance training to improve muscle power, while Straight et al. (2016) showed that resistance training mode moderated the exercise training effect on lower limb muscle power gains, with effect sizes being significantly greater for HVRT (0.62, 95% CI 0.42-0.82) than for traditional resistance training (0.20, 95% CI 0.04-0.35). In line with these previous studies, meta-

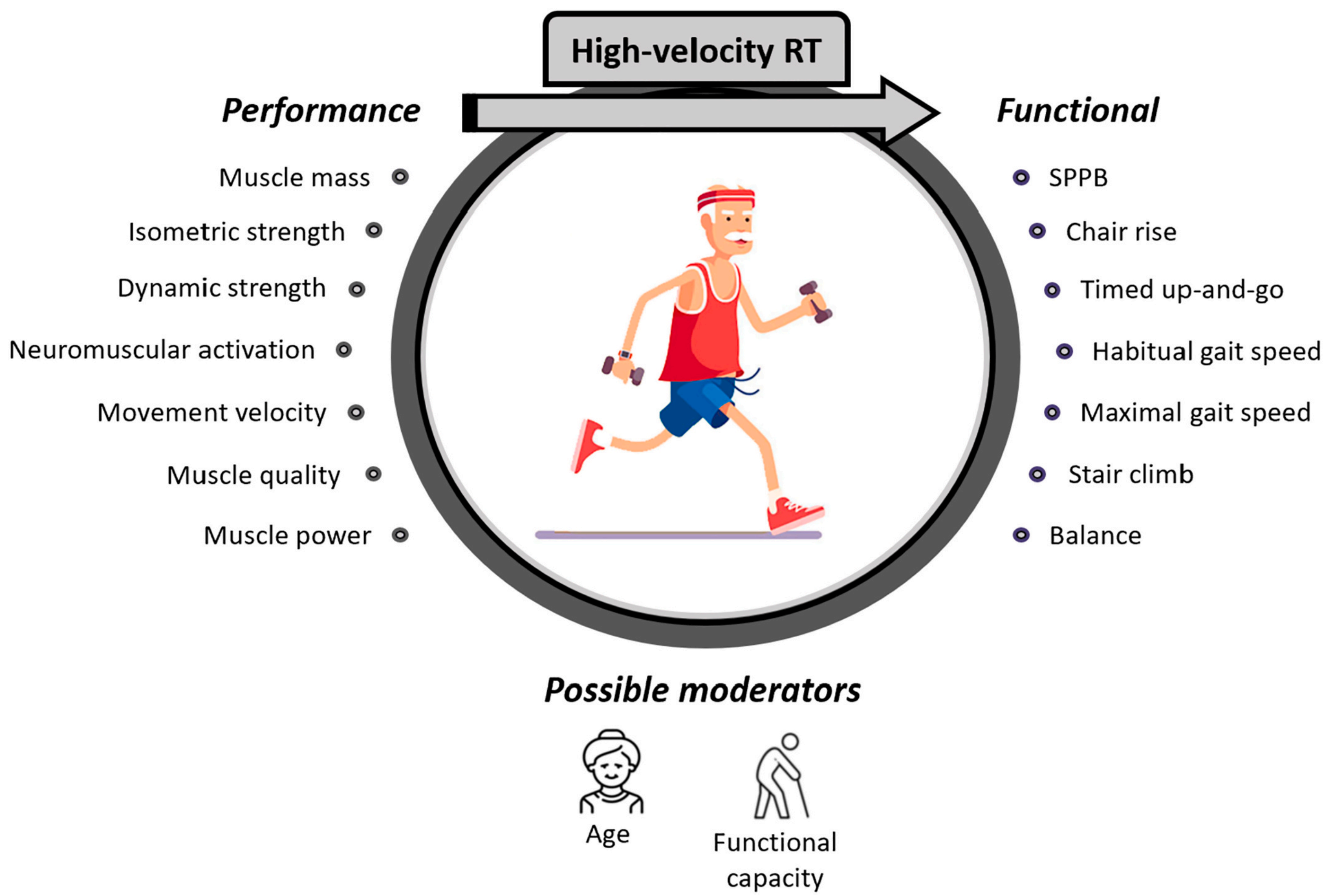

Fig. 2. Performance and functional outcomes that seem to be improved by high-velocity resistance training (RT) interventions in older adults. SPPB = *Short Physical Performance Battery*.

analyses of randomized clinical trials that directly compared both training modes also concluded that HVRT led to superior power and functional capacity adaptations (Steib et al., 2010; Tschopp et al., 2011). Most interestingly, these results in HVRT can be achieve using only low to moderate loads (40-60% 1RM) (Fragala et al., 2019), which may be advantageous for naïve exercisers and those with greater limitations. A summary of some known performance and functional outcomes improved by HVRT is illustrated in Fig. 2.

4.2. High-velocity resistance training in apparently healthy older adults

Initial investigations, mainly by Scandinavian researchers, helped to determine the importance of using explosive movements during resistance training (Häkkinen et al., 2000; Häkkinen et al., 1998a, 2001a; Häkkinen et al., 1998b, 2001b; Newton et al., 2002). Using a traditional resistance training program with a portion of its volume consisting of explosive exercises, these authors observed important increases in maximal and explosive strength, muscle cross-sectional area, neuromuscular activation, as well as in muscle power, in middle-aged and older men and women after periods between 16 and 24 weeks of training.

Among the first investigators to look into the responses of an exclusive HVRT program, Earles and collaborators (Earles et al., 2001) showed an ~20% increase in lower limb peak power and maximal dynamic strength after 12 weeks of training in apparently healthy older adults. These results were accompanied by improvements in the SPPB score and in the 6-min walk test performance. Indeed, several subsequent studies confirmed HVRT capacity to improve muscle power in elderly individuals (Balachandran et al., 2017; Bottaro et al., 2007; de Vos et al., 2005; Glenn et al., 2015; Henwood et al., 2008; Henwood and Taaffe, 2005; Radaelli et al., 2018; Sayers et al., 2012; Sayers and Gibson, 2014; Signorile et al., 2002; Tiggemann et al., 2016) and meta-analysis data observed a moderate effect size (0.62, 95% CI 0.42-0.82) of HVRT to improve leg press and knee extension power in individuals over 50 years (Straight et al., 2016).

In line with the muscle power results and its association with functional capacity discussed in the previous sections, several of the aforementioned studies also observed improvements in physical functioning. Henwood and Taaffe (2005) trained apparently healthy older adults for 8 weeks and found improvements in gait speed, chair rise and floor rise to stand tests. In a group of sedentary elderly people, increments in the 8 ft up and go, as well as in the sit-to-stand tests were also observed after 10 weeks of training (Bottaro et al., 2007). More recently, healthy older women improved their performance in both the TUG and stair climb tests after 12 weeks of HVRT (Radaelli et al., 2018). Again, these results are corroborated by several other studies (e.g., Balachandran et al., 2017; Correa et al., 2012; Henwood et al., 2008; Pereira et al., 2012; Ramirez-Campillo et al., 2014; Richardson et al., 2018), although it is worth mentioning that, in some cases, not all functional tests improved statistically (Dulac et al., 2018; Henwood and Taaffe, 2006). A summary of the studies identified that assessed power and functional performance outcomes in apparently healthy older adults are available in Table 1.

It is also important to highlight that in addition to adaptations in muscle power and functional performance, studies have also demonstrated improvements in other important outcomes. As an example, both maximal dynamic and isometric strength seem to respond positively to HVRT (Orssatto et al., 2018), however, different from muscle power gains which may be optimized with low to moderate loads (de Vos et al., 2005; Katsoulis et al., 2019; Reid et al., 2015), higher intensities seem to favor strength gains in HVRT (Byrne et al., 2016; Orssatto et al., 2018). Adaptations in muscle mass (i.e. hypertrophy), commonly associated with traditional resistance training, are also observed after HVRT (Conlon et al., 2017; Correa et al., 2012; Henwood et al., 2008; Radaelli et al., 2018; Reid et al., 2015). In this regard, even though functional performance seems more associated to muscle strength and power, low levels of muscle mass are found to be related with impairments in functional performance (Orssatto et al., 2018). Considering that muscle mass also has an important role as a secretory organ, interacting with a wide array of tissues in the human body (Pedersen, 2013), this adaptation should not be neglected in this population even if one's focus is on physical function. Other responses include improvement of qualitative aspects impaired during aging, such as muscle quality or specific strength, for example (Casas-Herrero et al., 2013; Radaelli et al., 2019).

4.3. High-velocity resistance training in mobility-limited older adults

According to the *International Classification of Functioning, Disability and Health* (World Health Organization, 2001), mobility comprises movements related to changes in body position (e.g., standing up), body's center of gravity (e.g., changing the support of one foot to the other when walking) and displacement (e.g., walking and climbing steps). In the United States, for example, limitations on the ability to perform tasks like the ones mentioned above, which will be referred to as mobility limitations, affect approximately one in four older individuals, reaching close to three quarters of those who live institutionalized (Bean et al., 2003; Jette and Branch, 1981). A British study noted further that the prevalence of middle-aged adults (50–64 years) who reported some kind of difficulty in walking 400 m was close to 20%, increasing considerably in older age groups (Gardener et al., 2006). This is important, as limitations in mobility can lead to the loss of independence of these individuals and are related to functional decline and an increased risk of disability, institutionalization and mortality (Guralnik et al., 1995).

The number of studies that investigated the effects of HVRT in older people with limited mobility is relatively smaller when compared to healthy individuals. A portion of these investigations, yet, did not employ an objective mobility limitation assessment, but rather included subjects who reported having limitations in activities such as walking certain distances, climbing a variable number of flights of stairs, among others (e.g., Fielding et al., 2002; Marsh et al., 2009). Self-report and perception of limitation are important factors (Sayers et al., 2004), however, they make it difficult to compare studies because participants may underestimate or overestimate their functional capacity (Reid et al., 2008). Thus, objective measures seem advantageous from a validity and reproducibility perspective (Guralnik et al., 1994; Reid et al., 2008). Whatever the case, both studies that considered mobility limitations by self-report or measured it objectively seem to indicate that HVRT is capable of increasing muscle power and functional performance in these individuals. Indeed, even older adults with neurodegenerative diseases that impact mobility and movement quality can tolerate and benefit from HVRT. Older individuals with Parkinson's disease, for example, have showed improvements in upper and lower limb bradykinesia, muscle power and functional capacity, and no adverse events, in response to HVRT (Lima and Rodrigues-de-Paula, 2012; Ni et al., 2016a, 2016b).

Among the studies that investigated subjects with self-reported limitation, Fielding and co-workers (Fielding et al., 2002) observed significant increases in the leg press and knee extension peak power after 16 weeks of HVRT. These results were accompanied by a reduction in the time to climb a flight of stairs and an improvement in a dynamic balance test, but no differences were found in the sit-to-stand and gait speed tests (Sayers et al., 2003). Marsh et al. (2009) also observed an increase in leg press peak power and knee extension coupled with a marked improvement in the SPPB score after 12 weeks. Sayers and Gibson (2010), in turn, measured peak power at 40, 50, 60, 70, 80 and 90% 1RM and found improvements over the entire intensity range analyzed. These authors also measured peak power at these same intensities but using workloads relative to the pre-intervention 1RM and significant increases were also found. This last result is quite interesting, as in many cases the overloads faced by older individuals during their various daily tasks do not change significantly through time (e.g., one's own body mass when rising from a chair).

Table 1
Summary of the power and functional performance outcomes in the high-velocity resistance training studies that investigated apparently health older adults.

Author	Study type (comparator)	Subjects	Weeks	Freq	Intensity	Volume	Rest	Changes in power	Changes in functional performance
Earles et al. (2001)	RCT (Walking group)	18 men and women (77 ± 5 years)	12	3	50–70% 1RM	2–3 × 10 reps	90 s	LP PP (22%); LP PP 40-70% 1RM (28-150%)	SPPB (7%); TC6 (5%); PPT (ns)
Signorile et al. (2002)	RCT (Control + PRT)	I9 women (68 ± 1 years)	12	3	3,14 and 4,73 rad/s	2 × 6–10 reps	120 s	KE AP 1.05 rad/s (ns), 3.14 rad/s (31%); 5.24 rad/s (38%)	–
Macaluso and De Vito (2003)	Non-RCT (Single arm)	31 women (68 ± 1 years)	16	3	40–80% 2RM	8 × 8–16 reps	120 s	LP PP (11-20%); CMJ (13-16%)	MGS (16%)
Henwood and Taaffe (2005)	Non-RCT (Control)	15 men and women (70 ± 7 years)	8	2	35, 55 and 75% 1RM	3 × 8 reps	–	KE PP 60°/s (17%); 120°/s (17%); KE AP 60°/s (33%); 120°/s (23%)	CR (10%); HGS (7%); MGS (ns); BGS (ns); Floor rise time (10%)
de Vos et al. (2005)	RCT (Control)	28 men and women (68 ± 5 years)	12	2	20% 1RM	3 × 8 reps	–	PP (14%)	–
		28 men and women (69 ± 6 years)			50% 1RM			PP (15%)	
		28 men and women (69 ± 6 years)			80% 1RM			PP (14%)	
Henwood and Taaffe (2006)	RCT (Control + PRT + Mixed)	23 men and women (71 ± 6 years)	8	2	45, 60 and 75% 1RM	3 × 8 reps	60 s	SC power (ns)	CR (12%); SC (8%); Floor rise time (ns); HGS (ns); MGS (ns); BGS (ns); 400 mW (ns)
Orr et al. (2006)	RCT (Control)	28 men and women (68 ± 5 years)	12	2	20% 1RM	3 × 8 reps	–	–	Balance (11%)
		28 men and women (69 ± 6 years)			50% 1RM				Balance (2%)
		28 men and women (69 ± 6 years)			80% 1RM				Balance (0.3%)
Bottaro et al. (2007)	RCT (PRT)	11 men and women (67 ± 6 years)	10	2	40–60%1RM	3 × 8–10 reps	90 s	LP PP (31%); PP bench press (37%)	8ftUG (15%); CR (43%)
Caserotti et al. (2008)	RCT (Control + Dif. age cohorts)	20 women (63 ± 2 years)	12	2	75–80% 1RM	4 × 8–10 reps	–	PP (12%); CMJ (4–5%)	–
		12 women (82 ± 3 years)						PP (28%); CMJ (6–9%)	
Henwood et al. (2008)	RCT (Control + PRT)	19 men and women (71 ± 1 years)	24	2	45, 60 and 75% 1RM	3 × 8 reps	60 s	PP (51%)	CR (13%); SC (7%); HGS (ns); MGS (4%); BGS (16%); 400 mW (8%)
Henwood and Taaffe (2008) *Retraining data from previous study*	RCT (PRT)	15 men and women (72 ± 2 years)	12	2	45, 60 and 75% 1RM	3 × 8 reps	60 s	PP (26%)	CR (ns); SC (ns); HGS (ns); MGS (ns); BGS (ns); 400 mW (ns)
Nogueira et al. (2009)	RCT (PRT)	11 men (67 ± 6 years)	10	2	40–60%	3 × 8 reps	90 s	LP PP (31%); CP PP (37%)	–
Claflin et al. (2011)	RCT (Control + Dif. age cohorts)	15 men and women (75 ± 4 years)	14	3	~120°/s–350°/s	3 × 10	–	Type II fiber peak power (12%)	–
Correa et al. (2012)	RCT (Control + PRT + Functional)	13 women (67 ± 5 years)	6	2	8–12 RM	3–4 × 8–12 reps	120 s	CMJ (8%)	CR (8%)
			12	3	40% 1RM				

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Table 1 (*continued*)

Author	Study type (comparator)	Subjects	Weeks	Freq	Intensity	Volume	Rest	Changes in power	Changes in functional performance
Sayers and Gibson (2012)	RCT (Control + PRT)	25 men and women (71 ± 7 years)				3 × 12–14 reps		LP PP 40–90% (191–229 W) (% was not possible to calculate)	Braking reaction time (ns); Breaking speed (15%)
Pereira et al. (2012)	RCT (Control)	28 women (63 ± 5 years)	12	3	40, 60 and 75% 1RM	3 × 4–12 reps	120 s	CMJ (40%); Medball Throw (17%)	CR (18%); MGS (14%); 8ftUG (16%)
Leszczak et al. (2013)	RCT (Eccentric training)	9 men and women (74 ± 9 years)	8	2	50% 1RM	3 × 8–12 reps	–	–	CR (17%); HGS (7%); 8ftUG (6%)
Ramirez-Campillo et al. (2014)	RCT (Control + PRT)	15 women (66 ± 4 years)	12	3	40, 60 and 75% 1RM	3 × 8 reps	60 s	CMJ (23%); Medball Throw (20%)	CR (21%); MGS (14%); 8ftUG (18%)
Sayers and Gibson (2014)	RCT (Control + PRT)	24 men and women (71 ± 7 years)	12	3	40% 1RM	3 × 12–14 reps	–	LP PP (26%)	–
Pamukoff et al. (2014)	RCT (PRT)	10 men and women (73 ± 4 years)	6	3	50% 1RM	3 × 8–10	120–180 s	LP PP (27%); KE PP (26%)	Frontal balance (15%); Lateral balance (26%)
Uematsu et al. (2014)	RCT (Control)	15 men and women (74 ± 4 years)	8	2	30–40% 1RM	3 × 10 reps	120 s	–	HGS (11%); MGS (18%)
Glenn et al. (2015)	RCT (Non-weighted)	30 men and women (78 ± 1 years)	20	2	70% 1RM	3 × 10 reps	–	PP (9%); AP (17%)	CR (20%); 8ftUG (19%); HGS (25%); TC6 (ns)
Tiggemann et al. (2016)	RCT (PRT)	15 women (65 ± 4 years)	12	2	~45–75% 1RM	2–3 × 8–15 reps	120 s	CMJ (28%); SJ (34%)	CR (12%); SC (19%); TUG (5%); TC6 (9%)
Lopes et al. (2016)	RCT (Control + PRT)	12 women (67 ± 7 years)	12	3	40-80% 1RM	3-4 × 8–10 reps	180 s	–	CR (22%); TUG (10%); TC6 (ns)
Ramirez-Campillo et al. (2016)	RCT (Control)	24 women (63 ± 7 years)	12	2	75% 1RM	2–3 × 8 reps	–	Medball Throw (19%)	CR (24%); 8ftUG (10%); MGS (9%)
								Medball Throw (15%)	CR (27%); 8ftUG (10%); MGS (8%)
				3	75% 1RM				
Englund et al. (2017)	RCT (PRT)	13 men and women (65 ± 2 years)	6	3	240°/s	3 × 8 reps	180 s	AP 75°/s (8%); AP 180°/s (14%); AP 240°/s (19%)	SPPB (3%); 8ftUG (17%)
Beijersbergen et al. (2017b)	RCT (Control)	15 men and women (73 ± 5 years)	10	3	40-60% 1RM	3 × 6–10 reps	–	KE PP 60°/s (26%); KE PP 120°/s (27%); KE PP 180°/s (23%); PP knee flexors at the same %'s (30–13%)	SC (ns); HGS (ns); MGS (6%); TC6 (ns)
Beijersbergen et al. (2017a)	RCT (Control)	15 men and women (73 ± 5 years)	10	3	40–60% 1RM	3 × 6–10 reps	–	Plantar flexors PP 20°/s (48%); 40°/s (40%); 60°/s (42%)	–
Balachandran et al. (2017)	RCT (Dif. training machines)	17 men and women (69 ± 5 years)	12	2	OMNI 6-8 (Plate-loaded)	3 × 8–10 reps	60 s	PP (31%); Medball Throw (6%)	CR (18-22%); HGS (ns); 8ftUG (9%); Gallon jug transfer (10%); Balance (ns)
		19 men and women (69 ± 5 years)			OMNI 6–8 (Pneumatic)			PP (19%); Medball Throw (ns)	CR (16-18%); HGS (11%); 8ftUG (9%); Gallon jug transfer (8%); Balance (13%)
Conlon et al. (2016, 2017)	RCT (Dif. periodization schemes)	10 men and women (70 ± 6 years)	22	3	10 RM (Non-periodized)	3 × 10 reps	90–120 s	CMJ (9%)	CR (25%); SC (11%)
		13 men and women (72 ± 5 years)			15, 10, 5 RM (Block periodization)			CMJ (5%)	CR (18%); SC (9%)
		10 men and							CR (20%); SC (10%)

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Table 1 (*continued*)

Author	Study type (comparator)	Subjects	Weeks	Freq	Intensity	Volume	Rest	Changes in power	Changes in functional performance
		women (71 ± 4 years)						CMJ (5%)	
					15. 10, 5 RM (Wave periodization)				
Ramirez-Campillo et al. (2017)	RCT (Control)	30 women (68 ± 5 years)	12	3	45, 60 and 75% 1RM (superv. 1:1)	3 × 8 reps	60 s	CMJ (18%); Medball Throw (15%)	CR (20%) MGS (11%); 8ftUG (14%)
								CMJ (1%); Medball Throw (1%)	
									CR (19%) MGS (3%); 8ftUG (11%)
		28 women (66 ± 4 years)			45, 60 and 75% 1RM (superv. 1:10)				
Ramirez-Campillo et al. (2018)	RCT (Control)	20 women (68 ± 5 years)	12	3	45,60 and 75% 1RM	3 × 8 reps	120 s	–	CR (14%); MGS (7%); 8ftUG (9%)
									CR (20%); MGS (15%); 8ftUG (15%)
		15 women (68 ± 5 years)					30 s (every 2 rep)		
Dulac et al. (2018)	Non-RCT (Dif. protein intake)	10 men - prot. < 1,1 g (73 ± 3 years)	12	3	50 e 80% 1RM	1–3 × 10 rep	60 s	–	CR (ns); HGS (ns); MGS (ns); TUG (12%); Balance (ns); HGS (ns); MGS (ns)
									CR (ns); HGS (10%); MGS (ns); TUG (8%); Balance (ns)
		11 men - prot. > 1,2 g (73 ± 3 years)							
Gray et al. (2018)	RCT (Control + PRT)	20 men and women (82 ± 6 years)	48	2	50%	3 × 10 rep	–	CR PP (ns)	CR (38%); 8ftUG (6%)
Radaelli et al. (2018)	RCT	13 women (65 ± 3 years)	12	2	30, 45 and 60% 1RM	1 × 8–12	180 s	CMJ (15%)	SC (4%); TUG (5%)
									SC (7%); TUG (4%)
						3 × 8–12		CMJ (16%)	
		13 women (66 ± 2 years)							
Richardson et al. (2018)	RCT(Control + PRT)	11 men and women (66 ± 5 years)	10	1	40% 1RM	3 × 14 rep	90 s	–	CR (13%); 8ftUG (ns); TC6 (ns)
									CR (10%); 8ftUG (8%); TC6 (7%)
		11 men and women (67 ± 3 years)		2					
Uematsu et al. (2018)	RCT (Control)	15 men and women (74 ± 3 years)	8	2	30–40% 1RM	3 × 10 rep	120 s	–	HGS (11%); MGS (18%);

RCT = randomized controlled trial; **PRT** = traditional progressive resistance training; **1RM** = one repetition maximum test; **RM** = intensity based on maximal repetitions; **LP** = *leg press*; **KE** = knee extension; **PP** = peak power; **AP** = average power; **CMJ** = countermovement jump; **SPPB** = *Short Physical Performance Battery*; **CR** = chair rise test and variations; **SC** = stair climb; **HGS** = habitual gait speed; **MGS** = maximal gait speed; **BGS** = backwards gait speed; **TUG** = *timed up and go*; **8ftUG** = *8 ft up and go*; **TC6** = 6-min walk test; **400 mW** = 400-m walk test; Note: when percentage changes were not presented in the original manuscripts they were estimated based on the pre-post values for the outcomes that showed significantly effects ($p < 0,05$); sample size correspond only to the participants in the high-velocity resistance training groups; different manuscripts from the same study were only included if novel results were reported.

Among the experimental studies that used objective measures to classify individuals as having mobility limitations, the most frequent criterion was a score equal to or lower than 9 points on the SPPB test (Balachandran et al., 2014; Bean et al., 2004, 2009; Reid et al., 2008, 2015). Habitual gait speed values in distances ranging from three (Hvid et al., 2016) to six meters were also employed (Hruda et al., 2003). In general, improvements in muscle power and functional performance were found using both high (~70-80% 1RM) (Hvid et al., 2016; Reid et al., 2008, 2015) and low loads (~40% 1RM) (Miszko et al., 2003; Reid et al., 2015) and even using one's own body weight, weighted vests and elastic bands (Bean et al., 2004, 2009; Hruda et al., 2003). After 12 weeks of training, older participants showed an increase in leg press and knee extension peak power measured at both 40 and 70% 1RM (Reid et al., 2008). In a later experiment, the same research group compared the effects of 12 weeks of HVRT between two groups that exercised using either 40% or 70% 1RM (Reid et al., 2015). They found no significant differences in muscle power gains between groups when evaluated at 40% and 70% 1RM, nor in SPPB increments.

Using a HVRT program composed of exercises such as chair raising, climbing steps and stair climbing with weighted vests as overload, results from Bean et al. (Bean et al., 2009) agree with previous studies. The participants increased muscle power at 70% 1RM by 10% and SPPB score by 20% after 16 weeks of training. Secondary analysis of the aforementioned study by Bean et al. (2009) divided the mobility-limited older adults into two groups, namely, those who achieved a clinically significant change in their SPPB score or habitual gait speed (defined as 1 point or 0.1 $m \cdot s^{-1}$, respectively) and those who did not. Using multivariate logistic regression models, the authors observed that changes in muscle power were significantly associated with clinically important increases in SPPB scores and gait speed, regardless of changes in muscle strength (Bean et al., 2010). Thus, HVRT seems to be a relevant strategy for improving the functional performance of older adults with limited mobility, as well as for maintaining the independence of these individuals, however, more information is still needed regarding the responses of these individuals to this training mode. Table 2 presents a list of the studies identified that investigated power and functional performance outcomes in mobility-limited older adults.

4.4. *Comparing the effects of high-velocity resistance training in different aging cohorts*

It is important to bear in mind that not all older individuals possess the same functional and neuromuscular capacities. Understanding the possible differences in the way these individuals responded to HVRT can, therefore, assist in the development of training programs more closely related to the specific needs of these individuals. Moreover, unlike older individuals who typically already present deficits in the parameters discussed so far (Alcazar et al., 2018), middle-aged individuals represent a portion of the population in which several of the age-related impairments begin to show a more accelerated decline (Kennis et al., 2014; Kozakai et al., 2000; Landi et al., 2018; Leyva et al., 2016) and, thus, represent an important target for intervention in order to prevent or mitigate the most significant losses that would occur later in life. Although literature investigating the effects of HVRT on healthy older adults and mobility-limited older adults is somewhat large (the reader is referred to Tables 1 and 2), not much attention has been given to middle-aged adults. In addition, investigations directly comparing HVRT adaptations between groups like these are lacking, which makes it difficult to compare how they respond to exercise and, most importantly, if the mechanisms behind these adaptations are similar.

Indeed, it has recently been suggested that the mechanisms involved in the reductions of muscle power may differ to some degree between healthy and mobility-limited older adults (Reid et al., 2014). Study by the same group had already demonstrated differences in the associations between lower limb muscle power and its determinants between middle-aged adults and older adults with and without mobility-limitations (Pojednic et al., 2012). According to the authors, these results point to the growing importance of contraction velocity for power production with age (Pojednic et al., 2012) and, collectively, allow us to suggest the possibility of differences in adaptations between these individuals when submitted to the same HVRT program or, at least, that these results could be determined by different mechanisms.

As already mentioned, only a limited number of investigations compared the adaptations to HVRT among these groups. Häkkinen et al. (1998a) identified significantly and similar improvements in isometric and maximal dynamic strength in middle-aged and healthy older men and women, who completed 24 weeks of high intensity resistance training with ~20% of the total volume comprised of exercises performed in an explosive manner. These results were accompanied by an increase in the knee extensors cross-sectional area, neuromuscular activation of these same muscles measured during isometric and dynamic knee extensions contractions and during a vertical jump test, rate of force development and vertical jump power output. Interestingly, they also reported reductions in the coactivation of the antagonist biceps femoris muscle in the older, but not in the middle-aged adults. Such findings were pioneering in demonstrating the maintenance of the capacity of older individuals to respond to the training stimulus when compared to a younger group (ACSM, 2009), in addition to highlighting the relevance of adaptations beyond muscle mass. However, these results do not allow one to determine the exclusive effects of HVRT in these individuals, its contributions to different strength expressions (e.g., different percentages of 1RM), nor the impact of these results on functional performance, since only 20% of the total training volume was performed explosively (i.e., as fast as possible).

Systematic review with meta-analysis analyzed the effects of resistance training on lower limb muscle power in middle-aged and older adults (Straight et al., 2016). When testing the effect of possible moderators on muscle power gains, the authors did not observe a significant age effect, reinforcing the notion that older individuals maintain the ability to gain muscle power. However, it is important to note that only 2 of the included studies evaluated individuals under 60 years of age. In a separate analysis of the power-load curves of their previous study, Izquierdo et al. (Izquierdo et al., 2001) evidenced that increments differed between the middle-aged and older adults' groups in some of the loads investigated. That is, while middle-aged individuals increased muscle power at intensities ranging from 0 to 70% of 1RM, older adults improved only between 15 and 60%. Considering that power produced at different loads (e.g., 40 and 70% 1RM) may be more or less associated with the performance in distinct functional tasks (Cuoco et al., 2004), the assessment of power adaptations in a wide spectrum of loads can also be important to understand the relationship between neuromuscular and functional adaptations and the differences between how these groups respond to training.

Through the analysis of the results in the studies presented in Tables 1 and 2, it is possible to suggest that there may be some differences in the adaptations of healthy and mobility-limited older adults to HVRT. When averaged, healthy older adults' studies demonstrate an ~20% increase in muscle power, which is lower than the almost 36% observed in the studies that trained older participants with mobility limitations. Of course, the differences between how power was measured in each study preclude a direct comparison between these percentages, but, nevertheless, they suggest that mobility-limited individuals may have a greater response to training than those without limitations. It is also important to highlight that only two studies in each group did not find significant increases in power, that is, 96% and 93% of the studies found positive adaptations on this outcome in older adults with and without mobility limitations, respectively.

On the other hand, the comparison of functional performance tests' results can offer a more adequate comprehensive perspective on the possible differences in responses between these two groups, especially because the protocols for most tests show minor to no changes across studies (e.g., SPPB protocol, for example, remains constant).

Table 2
Summary of the power and functional performance outcomes in the high-velocity resistance training studies that investigated mobility-limited older adults.

Author	Study type (Comparator)	Subjects	Week	Freq	Intensity	Volume	Rest	Changes in power	Changes in functional performance
Self-reported mobility limitation									
Fielding et al. (2002)	RCT (PRT)	15 women (73 ± 1 years)	16	3	70% 1RM	3 × 8–10 reps	–	LP PP (97%); KE PP (33%)	–
Sayers et al. (2003)	RCT (PRT)	15 women (73 ± 1 years)	16	3	70% 1RM	3 × 8 reps	–	–	CR (ns); SC (13%); HGS (ns); MGS (ns); Dynamic balance (5%)
Sayers (2007)	RCT (Control + PRT)	5 men and women (71 ± 6 years)	12	3	40% 1RM	3 × 12–14 reps	60 s	KE PP 40–90% 1RM (19–28%) KE FPP 40–90% (11–14%); VPP (3–18%)	–
Marsh et al. (2009)	RCT (Control + PRT)	12 men and women (77 ± 6 years)	12	3	70%1RM	3 × 8–10 reps	Circuit	LP PP (42%); KE PP (34%)	SPPB (10%); Lateral mobility (73%)
Sayers and Gibson (2010)	RCT (Control + PRT)	13 men (74 ± 6 years)	12	3	40% 1RM	3 × 12–14 reps	–	*Relative to post-exercise 1RM*: LP PP 40–90% (9–29%); FPP (20–23%); VPP (−7 a 7%)	–
								Relativo to pre-exercise 1RM: LP PP 40–90% (22–58%); FPP (5–8%); VPP (18–50%)	
Webber and Porter (2010)	RCT (Control)	17 women (77 [70–88])	12	3	80% 1RM	3 × 8–10 reps	120 s	Plantar flexors PP (17%); dorsal flexors PP (27%)	–
								Plantar flexors PP (12%); Dorsal flexores PP (29%)	
					Elastic bands				
		17 women (74 [70-88])							
Objectively-measured mobility limitation									
Miszko et al. (2003) CS-PFP = ~55	RCT (Control + PRT)	18 men and women (72 ± 7 years)	16	3	40% 1RM	3 × 6–8 reps	–	*Wingate* PP (8%), *Wingate* AP (6%)	CS-PFP (15%)
Bean et al. (2004) SPPB ≤9	RCT (Control)	10 women (69 ± 6 years)	12	2	Body weight + weighted vest	3 × 10 reps	60–120 s	LP PP 40–90% 1RM (12–36%)	SPPB (35%); CR (44%); HGS (14%); Unilateral balance (50%)
Reid et al. (2008) SPPB ≤ 9	RCT (Control + PRT)	23 men and women (73 ± 7 years)	12	3	70% 1RM	3 × 8 reps	–	LP PP 40–70% 1RM (62–66%); KE PP 40–70% 1RM (28–32%)	–
Bean et al. (2009) SPPB ≤ 9	RCT (PRT)	72 men and women (75 ± 7 years)	16	3	Body weight + weighted vest	2 × 10 reps	–	PP (10%)	SPPB (20%)
Balachandran et al. (2014) SPPB ≤ 9	RCT (PRT)	9 women (72 ± 8 years)	15	2	50–80% 1RM	3 × 10–12 reps	Circuit	LP PP (41%); CP PP (24%)	CR (15%); SPPB (20%);
Reid et al. (2015) SPPB ≤ 9	Non-RCT	25 women (78 ± 5 years)	12	3	40% 1RM	3 × 10 reps		PP 40 (34%) e 70% 1RM (32%); VPP 40 (18%) e 70% 1RM (15%)	SPPB (16%)
								PP 40 (42%) e 70% 1RM (42%); VPP 40 (25%) e 70% 1RM (21%)	
					70% 1RM				

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Table 2 (*continued*)

Author	Study type (Comparator)	Subjects	Week	Freq	Intensity	Volume	Rest	Changes in power	Changes in functional performance
Hvid et al. (2016) 3-m HGS < 0,9 m/s	RCT (Control)	27 women (78 ± 4 years) 16 men and women (82 ± 1 years)	12	2	70–80% 1RM	3 × 8–10 reps	–	–	SPPB (22%) MGS (9%)
Frail, pre-frail and/or institutionalized									
Hruda et al. (2003) 6-m HGS baixa	RCT (Control)	18 men and women (85 ± 5 years)	10	3	Body weight + elastic bands	3 × 8 reps	–	AP 180°/s (60%)	CR (66%); HGS (36%); 8ftUG (32%)
Iwamoto et al. (2009) Institutionalized	RCT (Control)	34 men and women (75 ± 6 years)	20	3	Body weight	3 × 10 reps	–	–	CR (19%); HGS (13%); TUG (24%); Unilateral balance (~100%)
Drey et al. (2012) Pre-frail (Fried)	RCT (Control + PRT)	24 men and women (78 [72-80] years)	12	2	BORG 10–16	2 × 6–15	120	PP CR (ns)	SPPB (8%)
Zech et al. (2012) Pre-frail (Fried)	RCT (Control + PRT)	18 men and women (77 ± 6 years)	12	2	BORG 12–16	2 × 6–15 reps	120	PP CR (ns)	SPPB (12%)
Cadore et al. (2014a) Frail (Fried)	RCT (Control)	11 men and women (93 ± 3 years)	12	2	40–60% 1RM	1 × 8–10 reps	–	PP 30 (97%) e 60% 1RM (117%)	CR (58%); HGS (ns); TUG (6%)
Cadore et al. (2014b) Frail (Fried) and cognitively impaired	Non-RCT (Single arm)	18 men and women (88 ± 5 years)	8	2	20-50% 1RM	2 × 8–12 reps	–	–	CR (ns); HGS (ns); TUG (28%); Balance (~167%)
Tan et al. (2018) Pre-frail and frail (Freid)	Non-RCT (Single arm)	8 men and women (74 ± 10 years)	12	2	Body weight + elastic bands	2 × 10 reps	–	–	SPPB (14%); CR (ns); TUG (9%)

RCT = randomized controlled trial; **PRT** = traditional progressive resistance training; **1RM** = one repetition maximum test; **LP** = *leg press*; **KE** = knee extension; **PP** = peak power; **AP** = average power; **SPPB** = *Short Physical Performance Battery*; **CR** = chair rise test and variations (5×, 10×, 30 s); **SC** = stair climb; **HGS** = habitual gait speed; **MGS** = maximal gait speed; **BGS** = backwards gait speed; **TUG** = *timed up and go*; **8ftUG** = *8 ft up and go*; **6**minW = 6-min walk test; **400 mW** = 400-m walk test; Note: when percentage changes were not presented in the original manuscripts they were estimated based on the pre-post values for the outcomes that showed significantly effects ($p < 0{,}05$); sample size correspond only to the participants in the high-velocity resistance training groups; studies that combined aerobic training with the high-velocity resistance training intervention were not included.

Considering only the tests that were applied in both the healthy and mobility-limited older adults, the latter group had greater improvements in the SPPB (20% vs. 5%), sit-to-stand (30% vs. 19%), TUG (17% vs. 7%) and 8 ft up and go (32 vs. 12%) tests when compared to the healthy older adults after HVRT, while closer results between the groups were observed for the stair climb (13% vs. 9%), habitual (14% vs. 11%) and maximal gait speed (9% vs. 11%) tests, respectively. Of note, all studies in both groups found an improvement in SPPB and TUG performance. For the sit-to-stand test, only two studies in the healthy and one study in the mobility-limited older adults did not show an increase in the test and just one study in each group found no improvements in the 8-feet up and go.

Collectively, these results suggest the possibility that mobility-limited older adults may improve power and functional performance to a greater extent than their healthy counterparts after a HVRT program. They are also in line with the principle of diminishing returns, as recently proposed in the context of mobility by Brahms et al. (Brahms et al., 2021). As argued by the authors, individuals with greater mobility would have to invest more time to induce adaptations compared to those with a lower mobility status. Notwithstanding, it is important to highlight that these results do not clarify if these greater adaptations are able to bring these mobility-limited older adults back to the function and/or performance levels of (untrained) healthy individuals, nor if the mechanisms related to these adaptations are the same. As an example, it may be that the first group has a higher anabolic resistance and, therefore, while healthy older individuals can improve power and functional performance through hypertrophy, mobility-limited older adults may require a different adaptation to reach the same benefits. Whatever the case it may be, what is clear is that the adaptations can contribute to increase the functional reserve of the less functional groups (and not just them) and, thereby, protect these individuals against future deleterious events, such as diseases, hospitalizations, etc.

Considering that only a very limited number of investigations compared the effects of a HVRT intervention in more than one of the aforementioned groups, differences in protocols (i.e. exercises, number of sets, training frequency, intensities used), in the way in which outcomes were assessed (e.g., equipment used, load or velocity employed), among other aspects, make it difficult to compare the results observed between the studies (Orssatto et al., 2018). As such, studies are needed comparing the adaptations of middle-aged adults and older adults of distinct functional capacities submitted to the same HVRT program, as well as determining the impact that these adaptations on functional performance or functional reserve. This information could help to better understand the effects of HVRT on aging-related processes and improve the prescription of this important training mode according to the needs of each of these individual groups. Such investigations could also contribute on the identification of the most responsive outcomes to training according to the analyzed group, contributing to the design of future studies.

5. Conclusion

Collectively, the available evidence suggests that high-velocity resistance training seems to be a safe and effective training paradigm for improving muscle power, functional performance, and mobility of older individuals. It also seems that mobility-limited older adults may improve power and functional performance to a greater extent than that of their healthy counterparts after a high-velocity resistance training program, which is in line with the principle of diminishing returns. Considering that only a very limited number of investigations directly compared the effects of HVRT intervention in more than one of the aforementioned groups, studies comparing the adaptations of middle-aged adults and older adults of distinct functional capacities submitted to comparable HVRT programs are needed to determine the differences in neuromuscular adaptations and their impact on functional performance and functional reserve in these individuals.

Funding

This study was financed in part by the Fundação de Amparo à Pesquisa do estado do Rio Grande do Sul (FAPERGS), Brazil – 19/2551-0001979-3.

CRediT authorship contribution statement

Gustavo Z. Schaun: Conceptualization, Methodology, Writing – Original Draft; *Cristine L. Alberton*: Conceptualization, Methodology, Writing – Review & Editing, Supervision; *Marcas M. Bamman*: Conceptualization, Writing – Review & Editing, Supervision. All authors read and approved the final manuscript.

Data availability

All data pertinent to the present review is included in the manuscript.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Acknowledgements

G.Z.S. is supported by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), whereas C.L.A is supported by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq).

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